

Battery Degradation and Frequency Stability Considerations in the Design of Off-Grid Mining Microgrids

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Abstract

This paper proposes a novel frequency-constrained microgrid (MG) planning model formulated as a Mixed-Integer Second-Order Cone Programming problem that optimally determines the capacity of renewable energy sources (RES), lithium-ion battery energy storage systems (BESS) and diesel generators while minimising total cost, including capital, operation and maintenance, emissions, and replacement costs as well as salvage value, over the lifetime of the MG. The model incorporates frequency requirements on the rate of change of frequency (RoCoF), frequency nadir, and steady-state frequency limits, using rotated second-order cone constraints based on a linear ramping model for both BESS and diesel generators. To accurately capture BESS degradation over the planning horizon, both calendar and cycling degradation are modelled and linearised using Special Ordered Set (SOS) constraints, ensuring optimal charge-discharge management. The planning framework also incorporates battery degradation to capture the long-term operational impacts on MGs. The proposed model is evaluated by a real-world Australian underground mining operation, demonstrating its ability to deliver a frequency-secure and cost-effective MG design. The findings show that the model increases diesel capacity by around 10% and deploys BESS in all scenarios to satisfy frequency requirements, whereas the baseline violates these in over 40% of scenarios.

Keywords: Off-grid microgrid, frequency stability, battery degradation, mixed-integer second-order cone programming, mining electrification

Nomenclature

A. Sets and Indices

b, B Index and set of battery storage systems

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e, E	Index and set of representative days
g, G	Index and set of diesel generators
i, Ψ	Index and set of energy sources
m, M	Index and set of daily average state of charge
n, N	Index and set of depth of discharge levels
t, T	Index and set of time intervals in a day
y, Y	Index and set of years

B. Parameters

α_{dp}, β_{dp}	linear fitting coefficients of salvage value vs relative capacity curve of the battery
$\Delta PL_{t,e,y}$	Power imbalance at time t , day e , and year y
Δt	Time interval duration
$\eta_b^{ch}, \eta_b^{dis}$	Battery b charging and discharging efficiencies
Λ, Γ	Penalty coefficients for load shedding and renewable curtailment
$\overline{P_{t,e,y}^{pv}}, \overline{P_{t,e,y}^{wt}}$	Forecasted normalised wind and solar production at time t , day e , and year y
π_e	Equivalent number of days represented by cluster e in a year
$\underline{c_b^{rate}}, \overline{c_b^{rate}}$	Minimum and maximum c-ratings of battery b
c^E	Emission factor
c_g^I	Unit capital investment cost for diesel generator g
c_g^r	Primary reserve cost of generator g
c_i^{om}	Operation and maintenance (O&M) cost coefficient for renewable unit i or battery b
c_b^d	Penalty factor of battery degradation based on energy throughput
c_g^f, c_g^n	Fuel and no-load cost factors of diesel generator g
c_{PV}^I, c_{WT}^I	Per unit capital cost of solar and wind systems

c_p^I, c_e^I	Per unit capital cost of power and energy components of battery
CL	Length of the interval for the discretisation of the cycle ageing
EoL_b	End-of-life capacity threshold for battery b
f^0	System nominal frequency
H_g	Inertia constant associated with generator g
ir	Compound interest rate
L_i	Lifetime of energy source i including diesel generator, solar and wind systems
L_{BT}^p	Battery power unit lifetime
P_i^{max}	Maximum allowed capacity of energy source i
$Pd_{t,e,y}$	Electric load demand at time t , day e , and year y
$RoCoF^{max}$	Maximum acceptable rate of change of frequency (RoCoF) value
RU_g, RD_g	Ramp-up and ramp-down limits of generator g
T_G, T_B	Reserve delivery times of diesel generators and batteries
T_{DB}	Deadband time of governors of the diesel generators
UT_g, DT_g	Minimum up-time and down-time constraints for generator g

C. Decision Variables

$\Delta f(\tau)$	Frequency deviation following a power disturbance
$\lambda_{b,y}$	Binary variable that is 1 if the battery b is replaced in year y , zero otherwise
$\vartheta_{b,t,e,y}$	Binary variable for cycle indication
$CB_{b,y}$	Capacity of battery b in year y
P_b^{BT}, E_b^{BT}	Installed power and energy capacity of battery b
P_g^{max}	Maximum installed capacity of diesel unit g
P_i	Rated capacity of installed renewable energy source i

$P_{b,t,e,y}^{dis}, P_{b,t,e,y}^{ch}$	Battery b discharge and charge power at time t , day e , and year y
$P_{g,t,e,y}$	Power generated by diesel unit g at time t , day e , and year y
$P_{t,e,y}^{cu}$	Renewable energy curtailed at time t , day e , and year y
$P_{t,e,y}^{ls}$	Unserved load at time t , day e , and year y
$P_{t,e,y}^{WT}, P_{t,e,y}^{PV}$	Power output from wind and solar photovoltaic (PV) units at time t , day e , and year y
$R_{b,t,e,y}^B$	Reserve capacity contributed by battery b at time t , day e , and year y
$R_{g,t,e,y}^G$	Primary reserve provided by generator g at time t , day e , and year y
$SoE_{b,0,e,y}$	Initial energy stored in battery b , day e , and year y
$SoE_{b,t,e,y}$	Energy stored in battery b at time t , day e , and year y
$T_{g,t,e,y}^{ON}, T_{g,t,e,y}^{OFF}$	Consecutive ON/OFF time duration of generator g at time t , day e , and year y
$u_{g,t,e,y}$	Binary variable indicating the operational status of generator g at time t , day e , and year y
$z_{b,t,e,y}$	Binary variable indicating charging/discharging mode of battery b at time t , day e , and year y

D. Abbreviations

BESS	Battery Energy Storage System
BLAST	Battery Lifetime Analysis and Simulation Tool
DoD	Depth of Discharge
ENS	Energy Not Served
EoL	End-of-Life
GHG	Greenhouse Gas
HPC	High-Performance Computing
MG	Microgrid
MILP	Mixed Integer Linear Programming
MISOCP	Mixed Integer Second-Order Cone Programming

NPC	Net Present Cost
NSGA-II	Non-dominated Sorting Genetic Algorithm-II
PFR	Primary Frequency Response
RES	Renewable Energy Sources
RoCoF	Rate of Change of Frequency
SoC	State of Charge
SoE	State of Energy
SOS	Special Ordered Set

1. Introduction

The mining industry is one of the most energy-intensive sectors, often operating in remote regions with limited access to conventional power grids [1]. In Australia, more than 35% of the mines rely on off-grid energy solutions, requiring standalone power systems to ensure operational reliability [2]. To reduce fossil fuel dependency and greenhouse gas (GHG) emissions, the industry is accelerating the adoption of renewable energy sources (RES) and battery energy storage systems (BESS) in mining microgrids (MGs), driven by environmental regulations and increasing sustainability demands [1, 3]. However, the integration of RES in off-grid mining MGs presents significant frequency stability challenges due to the predominance of power electronics-based generation. Unlike conventional synchronous generators, inverter-based RES and BESS contribute limited synthetic inertia and primary frequency response (PFR), making these MGs more susceptible to rapid frequency deviations after sudden power disturbances [4, 5]. The shift of mining to high RES MGs makes frequency stability in MG design a vital research and operational issue.

Ensuring frequency stability in MGs has been a growing research focus, with several studies incorporating frequency constraints into operational models to establish minimum inertia and PFR requirements. For instance, a mixed-integer linear programming (MILP) model was proposed in [5] to determine the optimal size of BESS in grid-connected MGs, where frequency stability constraints were formulated based on a discretised version of the swing equation. The study in [4] proposed a frequency-constrained MG sizing model incorporating wind turbines, hydrogen systems, and BESS, capturing both short- and long-term uncertainties via distributionally robust chance constraints. More recently, [6] proposed a chance-constrained optimal

sizing scheme for BESS that coordinates with synchronous generators and emergency load shedding to satisfy frequency stability requirements under uncertain renewable disturbances. Li et al. [7] developed a frequency-secured planning method for integrated electricity-heat MGs, explicitly coordinating heterogeneous PFR resources and virtual inertia suppliers to ensure dynamic and static frequency security across all operating conditions. Beyond operational-level studies, [8] extended frequency-constrained modelling to the planning stage for integrated power and gas systems, embedding both under- and over-frequency security requirements alongside a deep-learning-based clustering method for representative day selection. Similarly, [9] proposed a two-layer expansion planning model for wind-photovoltaic-hydro-thermal systems that linearises frequency security constraints via a boundary equivalence method, while [10] and [11] embedded linearised frequency nadir and post-disturbance constraints directly into MILP-based BESS planning and allocation models. [12] further incorporated distributionally robust treatment of renewable failures within a frequency-constrained expansion planning framework. While these studies confirm the growing recognition that frequency security must be addressed at the planning stage rather than left to post-hoc operational checks, none of them jointly models heterogeneous diesel and inverter-based PFR dynamics, nor do they incorporate battery degradation over the planning horizon. The studies in [13, 14] incorporated droop-based control strategies into MG design to enhance both frequency and voltage stability. Lastly, [15] proposed a BESS sizing approach for a remote MG on Flinders Island, Australia, solved via a grey wolf optimisation algorithm, with system frequency dynamics assessed using the DigSilent PowerFactory simulation platform.

Although the above studies offer valuable information on frequency constraints in MG planning, these frameworks and methodologies primarily address typical residential, commercial, and community MG setups, either grid-connected or off-grid. None of these studies specifically focus on the distinct challenges faced by the heavy industrial sectors, particularly remote mining operations. In mining environments, MGs are tasked with providing a substantial, high-quality, and often continuous energy supply, which is generally operational 24 hours a day, 7 days a week. This demand pattern presents considerable challenges for maintaining frequency stability, especially with the increasing integration of RES [16, 17]. For example, the large-scale incorporation of RES often reduces reliance on traditional synchronous generators that supply critical inertia and PFR, leading to a decline in system-wide inertia and making frequency regulation more challenging [18, 19]. Additionally, during peak generation hours, wind and solar technologies may produce surplus energy, leading to financial losses from the curtailment of excess power. Although some of this energy can be stored and reused by BESS, the short-duration capabilities of existing battery technologies make the large-scale storage cost prohibitive. Moreover, declining ore grades and the resulting need for more intensive

extraction processes increase energy consumption [17]. At the same time, the deployment of renewable energy infrastructure is constrained by limited land availability, which frequently overlaps with high-value mineral zones [16, 17]. A further distinguishing factor is the electrification of heavy-duty machinery, such as haul trucks and excavators, which require ultrafast chargers capable of delivering hundreds of kWh to several megawatt-hours in minutes. These high-power, short-duration charging events create spiky and unpredictable load profiles, inducing sharp frequency excursions that conventional control schemes struggle to manage. In MGs already challenged by low inertia and volatile RES output, such demand spikes exacerbate instability risks and highlight the need for advanced control strategies and planning frameworks specifically tailored to the operational dynamics of mining applications [20, 21].

Previous research underscores the vital function of BESS in mitigating frequency challenges in MGs, yet numerous planning models, e.g., [4, 5, 19, 22], neglect BESS degradation over time. Accurate modelling of battery degradation is essential for effective MG planning. Some studies estimate battery life using cycle life functions relative to the depth of discharge (DoD). For example, [23] linearised the DoD-cycle relationship to find an optimal DoD; however, this approach does not account for capacity degradation over time. In [24], a DoD regulation strategy was proposed for BESS planning in off-grid MGs to reduce cycling frequency, improve economic benefits, and extend battery lifetime. Ref.[25] assessed battery capacity annually, replacing batteries when 80% of the nominal capacity is reached. Other research efforts have overly simplified the battery degradation model in long-term planning studies. Ref. [26] treated battery degradation as a constant annual factor, while [27] restricted annual battery dispatch without modelling degradation factors. More advanced approaches, such as [28], distinguish between calendar and cycle ageing but assume constant cycling patterns, reducing accuracy in real-world applications. Seger et al. [29] proposed a Li-ion degradation model integrated into the optimal design and energy management of an isolated MG that ignoring storage degradation leads to cost underestimations exceeding 20%. The study by [30] introduced a MILP-based BESS sizing framework that integrates degradation functions, highlighting the necessity of considering degradation to enhance reliability and cost-efficiency. Although their degradation modelling offers a robust basis for incorporating battery ageing within an optimisation framework, it presumes a constant replacement cost and overlooks the salvage value of partially degraded batteries at the planning horizon's end. León et al. [31] presented an optimal design methodology for photovoltaic (PV) and hybrid-storage MGs in non-interconnected areas, further confirming that omitting battery degradation in planning leads to unreliable sizing and higher long-term costs. [32] proposed an iterative multi-year MILP framework for rural MG planning that captures power-dependent storage efficiency and capacity fade through an

external update loop, reducing computational burden relative to a full-horizon single-shot formulation. However, this framework does not incorporate frequency security constraints and targets a much smaller, single-battery rural MG context rather than the heterogeneous multi-asset, high-baseload conditions of remote mining operations.

In prior work [17], a multi-objective frequency-constrained MG planning framework was developed for the mining industry, where battery degradation was modelled using the Battery Lifetime Analysis and Simulation Tool (BLAST) library [33] and the complex planning problem was solved using evolutionary programming techniques such as the non-dominated sorting genetic algorithm-II (NSGA-II), enabling reliable operation under high renewable energy penetration. That framework relied on indirect necessity constraints to infer frequency feasibility under the assumption that the BESS is sufficiently fast to immediately compensate power imbalances, and therefore only explicitly modelled the frequency response of the diesel generators. While effective for limited configurations, this approach does not scale to heterogeneous inverter-based resources with distinct response dynamics and does not support a unified mathematical representation of multiple frequency-responsive units. Moreover, battery degradation was evaluated only over a single operational year, limiting the framework’s ability to capture long-term effects over the full planning horizon.

Despite the growing body of work reviewed above, a critical and unresolved gap remains in the literature. Existing frequency-constrained planning studies [4–12], including those that explicitly move frequency security from the operational to the planning stage [8, 9], focus on grid-connected or generic islanded/bulk power systems and do not account for the simultaneous long-term cost implications of BESS degradation. Conversely, degradation-aware planning models [28–31] omit frequency security constraints entirely, leaving inertia and PFR requirements unaddressed. The few studies that consider both frequency stability and battery degradation, including prior work by the authors [17], either rely on metaheuristic solvers that do not scale to heterogeneous multi-asset systems, use indirect feasibility screening instead of explicit frequency-constraint embedding, or model degradation only over a single representative year. No existing framework simultaneously: (i) explicitly represents the PFR dynamics of *heterogeneous* diesel and inverter-based resources with distinct ramping characteristics; (ii) embeds RoCoF, frequency nadir, and quasi-steady-state constraints directly as *mathematically rigorous* rotated second-order cone constraints within a long-term planning optimisation; and (iii) accounts for the *cumulative* impact of both calendar and cycling battery degradation over the full project lifetime in a tractable single-level formulation. This gap is especially critical in remote *mining* MGs, where 24/7 high-baseload operation, spiky ultrafast truck-charging events (capable of imposing step changes exceeding 10% of site demand in seconds), and inherently low system inertia

invalidate the simplifying assumptions routinely adopted in community or residential MG studies.

To address these limitations, this paper extends previous work [17] by developing a single-level mixed-integer second-order cone programming (MISOCP) formulation in which the frequency response of both synchronous diesel generators and inverter-interfaced resources is explicitly represented using linear ramping dynamics. This enables RoCoF, frequency nadir, and quasi-steady-state constraints to be enforced directly within the optimisation model. In addition, battery degradation is modelled over the entire planning horizon rather than a single representative year, allowing the long-term cost implications of operational decisions to be internalised. Compared with the earlier metaheuristic-based approach, the proposed formulation supports multiple heterogeneous frequency-responsive assets with distinct ramping characteristics and eliminates the need for indirect feasibility screening, thereby substantially improving scalability, solution consistency, and practical interpretability.

Table 1 summarises the existing literature and the proposed model in terms of MG configuration, application domain, problem focus, treatment of frequency requirements, and battery degradation modelling. The comparison visually corroborates the knowledge gap described above: no existing work occupies all three columns (off-grid, frequency constraints, battery degradation) while simultaneously employing a tractable mathematical programming formulation tailored to the industrial mining context.

The key contributions of this paper are as follows:

- **Mining-specific MG planning:** Development of a long-term planning framework tailored to remote mining MGs, which differ fundamentally from conventional residential or community MGs due to their 24/7 high baseload, spiky ultrafast charging loads from electrified heavy machinery, and limited system inertia — characteristics that invalidate assumptions adopted in prior frequency-constrained planning studies [4–7].
- **Integration of frequency stability in long-term planning:** Incorporation of all three frequency stability metrics—RoCoF, nadir, and steady-state frequency deviation—directly as rotated second-order cone constraints within the optimisation framework, ensuring robustness against frequency excursions caused by renewable intermittency and mining load spikes. Unlike prior degradation-aware planning studies [28–31], frequency security is an explicit design constraint, not a post-hoc check.
- **Degradation-aware storage modelling:** Integration of both cycle and calendar battery ageing mechanisms into the optimisation problem via SOS constraints, enabling accurate estimation of lifetime costs, optimal replacement schedules, and salvage value of BESS over the *full* planning horizon—

Table 1: Comparison of related frequency- and degradation-aware MG planning studies

Ref.	Off-grid MG	Frequency Constraints	Battery Degradation	Optimisation Model	Application / Remarks
[4, 5]	–	✓	–	MILP	Frequency-constrained planning for grid-connected MGs without degradation modelling or mining-specific operational characteristics.
[6]	–	✓	–	MILP	Chance-constrained BESS sizing for frequency stability with emergency load shedding; no degradation or off-grid/mining context.
[7]	–	✓	–	Evolutionary programming	Frequency-secured planning for integrated electricity-heat MGs with virtual inertia suppliers; no degradation modelling.
[8]	–	✓	–	MILP	Planning-level frequency security for integrated power-gas systems with deep-learning-based representative-day clustering; no degradation or off-grid/mining context.
[9]	–	✓	–	Two-layer MILP	Frequency-constrained wind/PV expansion planning via boundary-equivalence linearisation; no degradation modelling.
[10–12]	–	✓	–	MILP	Linearised frequency nadir/post-disturbance constraints embedded in BESS sizing and allocation; grid-connected, no degradation.
[32]	✓	–	✓	Iterative MILP	Multi-year rural MG planning with iterative storage efficiency/capacity-fade updates; no frequency security, small single-battery scale.
[13, 14]	–	✓	–	Evolutionary programming	Meta-heuristic-based frequency-constrained residential MG planning without lifetime battery degradation.
[15]	✓	✓	–	Evolutionary programming	Off-grid residential MG design considering frequency constraints but neglecting battery ageing and industrial-scale load disturbances.
[19, 22]	–	✓	–	MILP	Frequency-constrained battery sizing and unit commitment for bulk power systems without considering degradation.
[23, 27]	–	–	✓	MILP	Degradation-aware BESS sizing and scheduling in large power systems without frequency security.
[24]	✓	–	✓	Evolutionary programming	Off-grid MG sizing with battery degradation but no frequency-security modelling.
[25, 28, 30]	–	–	✓	MILP	Degradation-aware BESS planning for grid-connected power systems without frequency constraints.
[29]	✓	–	✓	MILP	Li-ion degradation model integrated into isolated MG design; omits frequency security constraints.
[31]	✓	–	✓	MILP	Optimal design of PV and hybrid-storage MGs with battery degradation for non-interconnected areas; no frequency constraints.
[26]	✓	–	✓	MILP	Off-grid MG planning considering degradation but assuming perfect frequency regulation.
[34]	✓	–	✓	Evolutionary programming	Mining MG sizing with degradation modelling using BLAST but neglecting frequency stability constraints.
[17]	✓	✓	✓	Evolutionary programming	First frequency- and degradation-aware mining MG planning study, but based on meta-heuristics with diesel-only frequency dynamics and indirect feasibility screening.
Proposed	✓	✓	✓	MISOCP	Unified frequency- and degradation-aware planning of off-grid mining MGs using a tractable single-level MISOCP formulation that explicitly models heterogeneous diesel and inverter-based frequency response over the full project lifetime.

extending beyond the single-year degradation assessment in [17].

- **Mathematically rigorous yet tractable formulation:** Formulation of the overall problem as a MISOCP, eliminating the need for metaheuristic solvers or indirect feasibility screening while retaining accuracy, scalability, and interpretability.
- **Practical insights for mining decarbonisation:** Application of the model to a real-world Australian underground mining case study, including rigorous sensitivity analyses across demand growth, frequency operating standards, and battery chemistry parameters, providing actionable guidance for mine operators, system designers, and regulators.

The rest of this paper is structured as follows: Section 2 details the problem formulation and frequency control mechanisms provided by BESS and diesel generators. Section 3 presents the linearisation approach used to transform the proposed formulation into a MISOCP problem. Section 4 discusses the numerical results and key findings, while Section 5 concludes the paper and outlines potential directions for future research.

2. Problem Formulation

This section presents the mathematical formulation for the proposed frequency-constrained planning problem, including the objective function, energy scheduling constraints, and frequency stability requirements. The frequency control mechanisms of BESS and diesel generators are modelled using a linear ramping approach to ensure reliable and resilient MG operation.

2.1. Objective Function

The objective function of the proposed planning model is to minimise the total net present cost (NPC), including capital cost (C^I), operating cost (C^O), replacement cost (C^R), emission cost (C^E), and salvage value (C^S) as follows:

$$\min NPC = C^I + C^O + C^R + C^E - C^S \quad (1)$$

where

$$\begin{aligned} C^I = & \sum_{g \in G} c_g^I P_g^{max} + c_{PV}^I P^{PV} + c_{WT}^I P^{WT} \\ & + \sum_{b \in B} \left(c_p^I P_b^{BT} + c_e^I E_b^{BT} \right) \end{aligned} \quad (2)$$

$$\begin{aligned}
C^O = & \sum_{y \in Y} \sum_{e \in E} \sum_{t \in T} \pi_e \left[\sum_{g \in G} \left(c_g^f P_{g,t,e,y} + c_g^n u_{g,t,e,y} + c_g^r R_{g,t,e,y}^G \right) \Delta t \right. \\
& + \sum_{b \in B} \left(c_b^d (-P_{b,t,e,y}^{ch} + P_{b,t,e,y}^{dis}) \Delta t + c_b^{om} P_b^{BT} \right) + c_{PV}^{om} P^{PV} \\
& \left. + c_{WT}^{om} P^{WT} + (\Gamma P_{t,e,y}^{cu} + \Lambda P_{t,e,y}^{ls}) \Delta t \right] \frac{1}{(1+ir)^y}
\end{aligned} \tag{3}$$

$$C^R = \sum_{y \in Y} \sum_{b \in B} \lambda_{b,y} \frac{c_e^I E_b^{BT}}{(1+ir)^{y-1}} + \sum_{n=1}^{\lfloor \frac{|Y|}{L_{BT}^p} \rfloor} \sum_{b \in B} \frac{c_p^I P_b^{BT}}{(1+ir)^{n L_{BT}^p}} \tag{4}$$

$$C^E = \sum_{y \in Y} \sum_{e \in E} \sum_{t \in T} \sum_{g \in G} \pi_e c^E \left(c_g^f P_{g,t,e,y} + c_g^n u_{g,t,e,y} \right) \Delta t \frac{1}{(1+ir)^y} \tag{5}$$

$$C^S = \sum_{i \in \Psi} C_i^S; \quad \Psi = \{DG, PV, WT, BT\} \tag{6}$$

$$C_{DG}^S = \sum_{g \in G} c_g^I P_g^{max} \left(1 - \frac{2}{L_{DG}} \right)^{|Y|} d \tag{7}$$

$$C_i^S = c_i^I P^i \left(1 - \frac{2}{L_i} \right)^{|Y|} d; \quad i = \{PV, WT\} \tag{8}$$

$$\begin{aligned}
C_{BT}^S = & \sum_{b \in B} \left[c_p^I P_b^{BT} \left(1 - \frac{2}{L_{BT}^{p,u}} \right)^{|Y|} + c_e^I \left(\alpha_{dp} C B_{b,|Y|} \right. \right. \\
& \left. \left. - \beta_{dp} E_b^{BT} \right) \right] d + \sum_{b \in B} \sum_{y \in Y} (\alpha_{dp} E O L_b - \beta_{dp}) \\
& c_e^I E_b^{BT} \lambda_{b,y} \frac{1}{(1+ir)^y}
\end{aligned} \tag{9}$$

where $|Y|$ represents the cardinal number of the set Y and $c^E = \frac{c_e^{DG} \gamma^{DG}}{c_t^{DG}}$ in which c_e^{DG} is the cost of emissions in \$/kg-CO₂, γ^{DG} is the emission production factor in kg CO₂/litre, and c_t^{DG} is the cost of diesel fuel in \$/litre. Furthermore, $d = \frac{1}{(1+ir)^{|Y|}}$ represents the discount rate at the end of the planning horizon and $L_{BT}^{p,u} = |Y| - \lfloor |Y|/L_{BT}^p \rfloor L_{BT}^p$ is the remaining useful life of the last battery unit b .

2.2. Technical constraints

The proposed model is subject to several constraints, which are detailed in the subsections below.

2.2.1. Constraints on planning capacity

Due to economic and technical considerations, the capacities of energy sources should be constrained as follows.

$$0 \leq P_i \leq P_i^{max}; \forall i \in \{PV, WT\} \tag{10}$$

$$0 \leq \sum_{g \in G} P_g^{max} \leq P_{DG}^{max} \quad (11)$$

$$0 \leq P_b^{BT} \leq P_b^{max} ; \forall b \quad (12)$$

$$\underline{C_b^{rate}} E_b^{BT} \leq P_b^{BT} \leq \overline{C_b^{rate}} E_b^{BT} ; \forall b \quad (13)$$

2.2.2. Normal Operation

The constraints on the normal operation of the MG are given below.

Supply-demand balance: This constraint guarantees the real-time balance between electricity demand and the combined supply from all available sources, as expressed below [17, 34]:

$$\begin{aligned} \sum_{g \in G} P_{g,t,e,y} + P_{t,e,y}^{PV} + P_{t,e,y}^{WT} + \sum_{b \in B} \left(P_{b,t,e,y}^{dis} + P_{b,t,e,y}^{ch} \right) \\ - P_{t,e,y}^{cu} + P_{t,e,y}^{ls} = P_{d,t,e,y} ; \forall t, e, y \end{aligned} \quad (14)$$

Renewable energy production: The output of the renewable energy units is determined by multiplying their installed capacity with the normalised forecast generation, as shown below [17].

$$P_{t,e,y}^{PV} = P_{PV} \overline{P_{t,e,y}^{pv}} ; \forall t, e, y \quad (15)$$

$$P_{t,e,y}^{WT} = P_{WT} \overline{P_{t,e,y}^{wt}} ; \forall t, e, y \quad (16)$$

Diesel generator constraints: The following constraints define the operational behaviour of diesel generators [17, 34].

$$0 \leq P_{g,t,e,y} \leq P_g^{max} u_{g,t,e,y} ; \forall g, t, e, y \quad (17)$$

$$-RD_g \leq P_{g,t,e,y} - P_{g,t-1,e,y} \leq RU_g ; \forall g, t, e, y \quad (18)$$

$$T_{g,t,e,y}^{ON} \geq UT_g (u_{g,t,e,y} - u_{g,t-1,e,y}) \Delta t ; \forall g, t, e, y \quad (19)$$

$$T_{g,t,e,y}^{OFF} \geq DT_g (u_{g,t-1,e,y} - u_{g,t,e,y}) \Delta t ; \forall g, t, e, y \quad (20)$$

where (17) and (18) limit the maximum/minimum production and the ramp-up/down of the diesel generators, while (19) and (20) denote the minimum up/down times of the generators.

BESS operation constraints: The following constraints govern the operation of BESS, in terms of charging/discharging limits (22) and (21), state of energy (SoE) calculation (23), and initial and end-of-day

SoE coordination (24) [17].

$$0 \leq P_{b,t,e,y}^{dis} \leq P_b^{BT} z_{b,t,e,y} ; \forall b, t, e, y \quad (21)$$

$$-P_b^{BT} (1 - z_{b,t,e,y}) \leq P_{b,t,e,y}^{ch} \leq 0 ; \forall b, t, e, y \quad (22)$$

$$SoE_{b,t,e,y} = SoE_{b,t-1,e,y} - \Delta t (\eta_b^{ch} P_{b,t,e,y}^{ch} + \frac{P_{b,t,e,y}^{dis}}{\eta_b^{dis}}) ; \forall b, t, e, y \quad (23)$$

$$SoE_{b,0,e,y} = SoE_{b,T,e,y} ; \forall b, e, y \quad (24)$$

Battery degradation: In this work, battery degradation over time is formulated to determine the replacement year and the capacity loss during operation. Battery degradation is generally classified into two categories: cycle ageing and calendar ageing. Cycle ageing is influenced by factors such as the number of charge-discharge cycles, the depth of discharge (DoD), the ambient temperature, the average state of charge (SoC) during each cycle, and the C-rate. Calendar ageing is mainly influenced by the average SoC, the duration of idle periods, and temperature. In critical applications, where battery temperature is regulated by active cooling and heating, the temperature's influence on degradation is largely negligible [28, 30]. Additionally, studies have shown that the stress effect caused by the average SoC has a minimal influence on cycle ageing [28, 35]. Consequently, cycle ageing can be reasonably estimated by considering the number of charge-discharge cycles and their depth, whereas calendar ageing is modelled using the average SoC and elapsed time (see Section 4.3 for a discussion of temperature, C-rate, and dwell-time effects on this modelling choice). The corresponding mathematical representations for battery degradation per cycle and calendar ageing per day are provided as follows [28, 30].

$$f^{cyc}(DoD_b) = A_b^{cyc} DoD_b^2 + B_b^{cyc} DoD_b \quad (25)$$

$$f^{cal}(SoC_b^{avg}) = A_b^{cal} (SoC_b^{avg})^2 + B_b^{cal} SoC_b^{avg} + C_b^{cal} \quad (26)$$

The coefficients for these equations, corresponding to four lithium-ion battery chemistries, namely nickel-manganese-cobalt (NMC), lithium iron phosphate (LFP), lithium manganese oxide (LMO), and lithium titanium oxide (LTO), are provided in [28]. These coefficients were derived from the experimental data presented in [28, 35]. Thus, the available battery capacity is determined by (27) and is constrained by (28), which is updated annually by deducting the capacity loss from battery degradation from the previous year's

usable capacity, as follows.

$$CB_{b,y} = E_b^{BT} \lambda_{b,y} + CB_{b,y-1} (1 - \lambda_{b,y}) - \sum_{e \in E} \pi_e \left(f^{cal}(SoC_b^{avg}) + \sum_{t \in T} \vartheta_{b,t,e,y} f^{cyc}(DoD_b) \right) E_b^{BT} ; \forall b, y \quad (27)$$

$$CB_{b,y} \geq EOL_b E_b^{BT} ; \forall b, y \quad (28)$$

where the cycle indicator $\vartheta_{b,t,e,y}$ is calculated based on the binary variable of charge and discharge status as given below.

$$\vartheta_{b,t,e,y} = (z_{b,t-1,e,y} - z_{b,t,e,y}) z_{b,t-1,e,y} ; \forall b, t, e, y \quad (29)$$

In addition, the SoE of the battery cannot exceed the maximum available capacity and cannot be discharged beyond the optimal DoD (DoD^{opt}) as follows [30].

$$CB_{b,y} (1 - DoD^{opt}) \leq SoE_{b,t,e,y} \leq CB_{b,y} ; \forall b, t, e, y \quad (30)$$

2.2.3. Frequency Dynamics Constraints

Consider a mining MG equipped with diesel generators, wind and solar systems, and batteries. Following a power disturbance, such as sudden loss of a generator or a surge of load, the subsequent frequency deviation $\Delta f(\tau)$ can generally be approximated using the swing equation $\forall t, e, y$, as given in [5]:

$$\frac{d\Delta f(\tau)}{d\tau} = \frac{1}{2H_{t,e,y}} \left(\sum_{g \in G} \Delta P_{g,t,e,y}^G(\tau) + \sum_{b \in B} \Delta P_{b,t,e,y}^B(\tau) - \Delta PL_{t,e,y} \right) \quad (31)$$

Note that since the off-grid MG relies mainly on power electronics and exhibits low inertia, the load damping effect is minimal [36] and is thus omitted from equation (31). In (31), $H_{t,e,y}$ is the aggregated inertia of the spinning diesel generators, which is expressed by:

$$H_{t,e,y} = \frac{\sum_{g \in G} H_g P_g^{max} u_{g,t,e,y}}{f^0} \quad (32)$$

In (31), PFRs provided by diesel generators and batteries ($\Delta P_{g,t,e,y}^G(\tau)$ and $\Delta P_{b,t,e,y}^B(\tau)$) are approxi-

mated using a linear ramping model, given below.

$$\Delta P_{g,t,e,y}^G(\tau) = \begin{cases} 0, & 0 \leq \tau \leq T_{DB} \\ R_{g,t,e,y}^G \frac{\tau - T_{DB}}{T_G}, & T_{DB} \leq \tau \leq T_G + T_{DB} \\ R_{g,t,e,y}^G, & \tau \geq T_G + T_{DB} \end{cases} \quad (33)$$

$$\Delta P_{b,t,e,y}^B(\tau) = \begin{cases} R_{b,t,e,y}^B \frac{\tau}{T_B}, & 0 \leq \tau \leq T_B \\ R_{b,t,e,y}^B, & \tau \geq T_B \end{cases} \quad (34)$$

Therefore, the constraints on the frequency requirements, including limitations on RoCoF, quasi-steady state, and frequency nadir, can be formulated as follows.

Constraint on RoCoF: The maximum RoCoF occurs immediately after a power disturbance, before any frequency response is provided. Thus, the RoCoF can be calculated by setting $\tau = 0$ in (31) and should be limited to a maximum allowable value.

$$H_{t,e,y} \geq \left| \frac{\Delta PL_{t,e,y}}{2RoCoF^{max}} \right| ; \forall t, e, y \quad (35)$$

Constraint on quasi-steady state: To ensure that the frequency decline is mitigated and eventually stabilises at a steady state, combined PFRs must be sufficient to counteract power disturbance. Thus, the following condition must hold:

$$\sum_{g \in G} R_{g,t,e,y}^G + \sum_{b \in B} R_{b,t,e,y}^B \geq \Delta PL_{t,e,y} ; \forall t, e, y \quad (36)$$

Constraint on frequency nadir: The frequency nadir is reached when the total PFR has fully counteracted the power disturbance. Due to the distinct delivery characteristics of the PFRs, which have different response times, the total PFR forms a piecewise linear function. Without loss of generality, it is assumed that $T_{DB} \leq T_B \leq T_G$. The piecewise linear total PFR is defined over the intervals $(0, T_{DB}]$, $(T_{DB}, T_B]$, $(T_B, T_G]$, and (T_G, ∞) . However, the frequency nadir must occur before T_G to prevent instability based on the quasi-steady state constraint (36). Thus, the frequency nadir will occur within one of the intervals $(0, T_G]$. For illustration, the interval $(T_{DB}, T_B]$ is considered where the frequency nadir occurs, and derive the corresponding constraint for the frequency nadir. By substituting (33) and (34) for $(T_{DB}, T_B]$ in

(31), the solution to (31) is:

$$\Delta f(\tau) = \frac{1}{2H_{t,e,y}} \left[\left(\frac{R_{t,e,y}^G}{2T_G} + \frac{R_{t,e,y}^B}{2T_B} \right) \tau^2 - \left(\Delta PL_{t,e,y} + \frac{R_{t,e,y}^G}{T_G} T_{DB} \right) \tau + \frac{R_{t,e,y}^G}{2T_G} T_{DB}^2 \right] ; \forall t, e, y \quad (37)$$

where $R_{t,e,y}^G = \sum_{g \in G} R_{g,t,e,y}^G$ and $R_{t,e,y}^B = \sum_{b \in B} R_{b,t,e,y}^B$. By setting $d\Delta f(\tau)/d\tau = 0$, the time to reach the frequency nadir τ^* is calculated by:

$$\tau^* = \frac{\Delta PL_{t,e,y} + R_{t,e,y}^G T_{DB}/T_G}{R_{t,e,y}^G/T_G + R_{t,e,y}^B/T_B} ; \forall t, e, y \quad (38)$$

By substituting τ^* values into associated $\Delta f(\tau)$, the frequency deviation at nadir $\forall t, e, y$ is calculated by:

$$\Delta f(\tau^*) = \frac{-\left(\Delta PL_{t,e,y} + R_{t,e,y}^G T_{DB}/T_G\right)^2}{4H_{t,e,y} \left(R_{t,e,y}^G/T_G + R_{t,e,y}^B/T_B\right)} + \frac{R_{t,e,y}^G T_{DB}^2}{4H_{t,e,y} T_G} \quad (39)$$

Given the maximum frequency deviation allowable Δf^{max} , the constraint on the frequency nadir can be achieved by:

$$|\Delta f(\tau^*)| \leq \Delta f^{max} \quad (40)$$

Thus, the full presentation of the constraint on the frequency nadir is obtained as follows for the period $(T_{DB}, T_B]$:

$$\left(\Delta PL_{t,e,y} + R_{t,e,y}^G \frac{T_{DB}}{T_G} \right)^2 \leq \left(\frac{R_{t,e,y}^G}{T_G} + \frac{R_{t,e,y}^B}{T_B} \right) \left(4H_{t,e,y} \Delta f^{max} + R_{t,e,y}^G \frac{T_{DB}^2}{T_G} \right) ; \forall t, e, y \quad (41)$$

Similarly, the full presentation of the frequency nadir constraints $\forall t, e, y$ for the periods of $(0, T_{DB}]$ and $(T_B, T_G]$ are:

$$\left(\Delta PL_{t,e,y} \right)^2 \leq \left(\frac{R_{t,e,y}^B}{T_B} \right) \cdot \left(4H_{t,e,y} \Delta f^{max} \right) ; (0, T_{DB}] \quad (42)$$

$$\left(\Delta PL_{t,e,y} + R_{t,e,y}^G \frac{T_{DB}}{T_G} - R_{t,e,y}^B \right)^2 \leq \left(\frac{R_{t,e,y}^G}{T_G} \right) \left(4H_{t,e,y} \Delta f^{max} + R_{t,e,y}^G \frac{T_{DB}^2}{T_G} - R_{t,e,y}^B T_B \right) ; (T_B, T_G] \quad (43)$$

Constraints on Primary Reserve: The primary reserves of BESS and diesel generators must be

constrained based on their installed capacities and power generation levels at each time interval, as follows.

$$0 \leq P_{g,t,e,y} + R_{g,t,e,y}^G \leq P_g^{max} u_{g,t,e,y} ; \forall g, t, e, y \quad (44)$$

$$0 \leq P_{b,t,e,y}^{dis} + R_{b,t,e,y}^B \leq P_b^{BT} z_{b,t,e,y} ; \forall b, t, e, y \quad (45)$$

$$-P_b^{BT}(1 - z_{b,t,e,y}) \leq P_{b,t,e,y}^{ch} - R_{b,t,e,y}^B \leq 0 ; \forall b, t, e, y \quad (46)$$

$$R_{b,t,e,y}^B, R_{g,t,e,y}^G \geq 0 \quad (47)$$

These constraints guarantee that the primary reserve assigned to each generator remains within its available capacity margin. Incorporating these constraints renders constraints (17), (22), and (21) unnecessary.

3. Linearisation and Problem Reformulation

The proposed optimisation problem is non-linear due to: (I) multiplication of decision variables, including binary-to-binary multiplications in (29) and binary-to-continuous interactions such as $\lambda_{b,y} \cdot E_b^{BT}$, $\lambda_{b,y} \cdot CB_{b,y}$, $P_g^{max} \cdot u_{g,t,e,y}$, and $P_b^{BT} \cdot z_{b,t,e,y}$ in (4), (9), (27), (17), (32), (22), (21), (45), and (46); and (II) the non-linearity in the calendar and cyclic ageing functions in (26) and (25), which are used to calculate the capacity of the BESS in each year (27).

To linearise the first group, the big-M method is employed in this paper [37]. Furthermore, Eqs. (25) and (26) are linearised using N and M discrete segments, respectively, as schematically shown in Figure 1. Then, new binary variables $W_{b,n,t,e,y}$ and $X_{b,m,e,y}$ are introduced for each segment, corresponding to specific degradation values, $degcyc_{b,n}$ and $degcal_{b,m}$, respectively, ensuring that only one binary variable is selected for each segment m or n . Therefore, Eq. (27) is updated to (48) by replacing the binary variables as:

$$\begin{aligned} CB_{b,y} &= E_b^{BT} \lambda_{b,y} + CB_{b,y-1} (1 - \lambda_{b,y}) \\ &\quad - \sum_{e \in E} \pi_e \sum_{m \in M} X_{b,m,e,y} degcal_{b,m} E_b^{BT} \\ &\quad - \sum_{e \in E} \pi_e \sum_{t \in T} \sum_{n \in N} W_{b,n,t,e,y} degcyc_{b,n} \vartheta_{b,t,e,y} E_b^{BT} ; \forall b, y \end{aligned} \quad (48)$$

Also, using this linearisation, the lower and upper bounds of SoE in (30) $\forall b, t, e, y$ are updated as:

$$SoE_{b,t,e,y} \leq CB_{b,y} CL + \sum_{n \in N} (1 - DoD_n) W_{b,n,t,e,y} CB_{b,y} \quad (49)$$

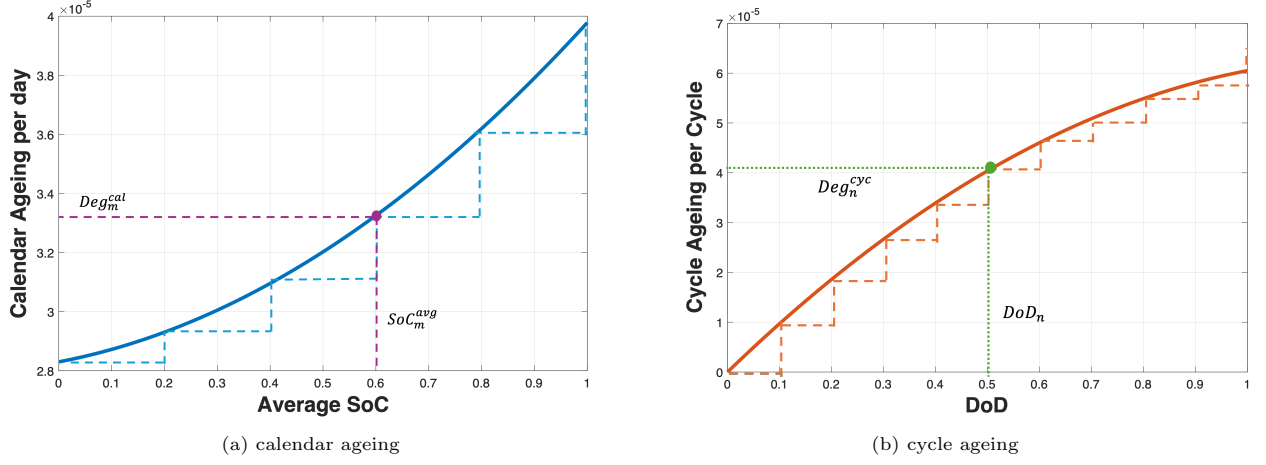


Figure 1: Schematic illustration of how calendar/cycle ageing is approximated across different segments.

$$SoE_{b,t,e,y} \geq \sum_{n \in N} (1 - DoD_n) W_{b,n,t,e,y} CB_{b,y} \quad (50)$$

where DoD_n represents the amount of DoD corresponding to $W_{b,n,t,e,y}$. Furthermore, since calendar ageing is related to the average SoC, the daily average SoC must not exceed the optimal average SoC for each day, as enforced by equation (51), where SoC_m^{avg} is the average SoC corresponding to $X_{b,m,e,y}$.

$$\frac{1}{|T|} \sum_{t \in T} SoE_{b,t,e,y} \leq \sum_{m \in M} X_{b,m,e,y} SoC_m^{avg} CB_{b,y} ; \forall b, e, y \quad (51)$$

Moreover, based on the SOS method, the battery can only be in calendar or cyclic ageing state at any time:

$$\sum_{n \in N} W_{b,n,t,e,y} = 1 ; \forall b, t, e, y \quad (52)$$

$$\sum_{m \in M} X_{b,m,e,y} = 1 ; \forall b, e, y \quad (53)$$

To this end, the calendar and cyclic ageing functions are linearised. However, (48), (49), (50), and (51) are still non-linear due to the binary-continuous multiplication, including $X_{b,m,e,y} \cdot E_b^{BT}$, $X_{b,m,e,y} \cdot CB_{b,y}$, and $W_{b,n,t,e,y} \cdot CB_{b,y}$, as well as the product of binary-to-binary-to-continuous terms, such as $W_{b,n,t,e,y} \cdot \vartheta_{b,t,e,y} \cdot E_b^{BT}$. These non-linear terms can be linearised using the big-M method [30, 37].

Constraints (41)-(43) represent rotated second-order cone constraints. Although these constraints are convex, they are defined over different intervals, and the frequency nadir will occur in one of these intervals. As a result, these constraints function as *if-then* conditional second-order cone constraints. To address this, three groups of binary variables, $z_{d,t,e,y}$ are introduced, where $d \in D = 1, 2, 3$ represents the interval index,

to indicate in which interval the frequency nadir occurs. For clarity, three bracketed terms are defined as $A_{d,t,e,y}(x)$, $B_{d,t,e,y}(x)$, and $C_{d,t,e,y}(x)$, where x represents the vector of optimisation variables. Using the big-M method and variable transformation, the *if-then* conditional constraints can be reformulated as the following constraints, incorporating new auxiliary variables $D_{t,e,y}$, $K_{t,e,y}$, and $L_{t,e,y}$, along with a predefined large positive constant, $bigM$, for all t, e, y :

$$D_{t,e,y}^2 \leq K_{t,e,y} L_{t,e,y} \quad (54)$$

$$\sum_{d \in D} z_{d,t,e,y} = 1 \quad (55)$$

$$-bigM(1 - z_{d,t,e,y}) \leq D_{t,e,y} - A_{d,t,e,y}(x) \leq bigM(1 - z_{d,t,e,y}) ; \forall d \in D \quad (56)$$

$$-bigM(1 - z_{d,t,e,y}) \leq K_{t,e,y} - B_{d,t,e,y}(x) \leq bigM(1 - z_{d,t,e,y}) ; \forall d \in D \quad (57)$$

$$-bigM(1 - z_{d,t,e,y}) \leq L_{t,e,y} - C_{d,t,e,y}(x) \leq bigM(1 - z_{d,t,e,y}) ; \forall d \in D \quad (58)$$

In summary, the full optimisation problem can be expressed concisely as follows:

$$\min \text{ Total NPC} \quad (1)$$

subject to

Planning constraints: (2)–(13),

Normal operation constraints: (14)–(24), (28)–(30), (48)–(53),

Frequency security constraints: (32), (35)–(36), (44)–(47), (54)–(58).

4. Simulation Results

This section presents the results of the proposed frequency-constrained and degradation-aware MG design model, applied to a real-world off-grid underground mining operation in Australia. The mine is located in a remote region with a considerable solar and wind energy potential, exhibiting capacity factors of approximately 43% for wind and 28% for solar. It operates continuously with a high and inflexible electrical load, characterised by a peak demand of 11.79 MW, an average load of 9.75 MW, and an annual energy consumption of 84.7 GWh, necessitating a stable and reliable power supply. These characteristics are illustrated in Figure 3, which shows the electricity load duration curve of the site together with the normalised wind and solar production duration curves [17, 34]. To aid readers in understanding the physical system

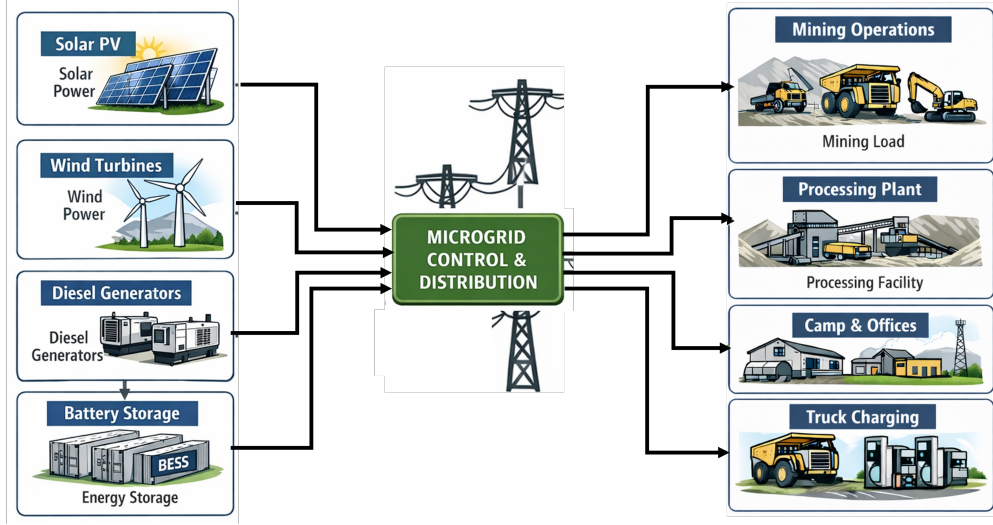


Figure 2: Simplified schematic of the mining MG studied in this paper.

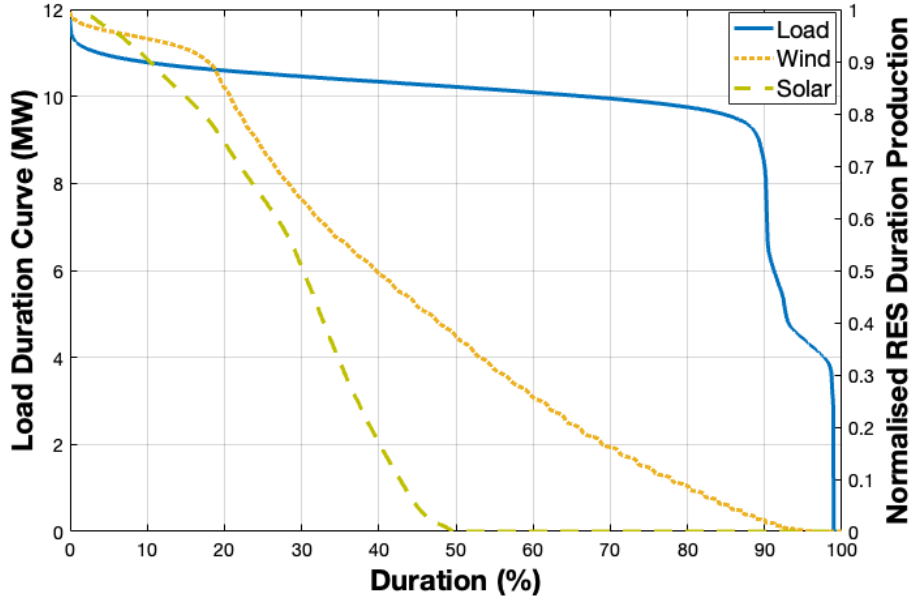


Figure 3: Electricity load duration curve and normalised potential RES production duration curve at the mine site.

being modelled, a simplified schematic of the test MG—including diesel generators, solar PV, wind turbines, BESS, and the mining loads—is provided in Figure 2. These operational demands introduce both technical and economic challenges in developing a (near) zero-emission MG system.

The simulation setup specifies the economic and technical parameters for each energy source, with all cost values expressed in Australian dollars [17, 34]. Diesel generators are selected as the fossil fuel source due to the lack of a natural gas pipeline at the site. Capital costs are set at \$1.8 million/MW for solar

photovoltaic (PV), \$2.2 million/MW for wind systems, and \$1.2 million/MW for diesel generators. For BESS, capital costs are \$0.187 million/MW and \$0.683 million/MWh. Operation and maintenance (O&M) costs are \$45/kW-yr for solar PV, \$60/kW-yr for wind systems, and \$7.57/kW-yr for BESS along with a penalty of \$1.00/kWh to avoid battery abuse due to unnecessary charge and discharge. Diesel generators incur O&M costs of \$0.171/kWh and \$14.92/h. Emission costs apply only to diesel generators, with a carbon price of \$0.07/kg-CO₂ and an emission factor of 3.3775 kg-CO₂/liter of fuel. The compound interest rate ir and the price of diesel fuel are assumed to be 3.1% and \$0.85/liter, respectively [38]. The assumed lifetimes are 25 years for solar PV, wind systems, and diesel generators, while the BESS power unit has a 10-year lifespan. The actual BESS energy capacity is updated annually with an end-of-life (EoL) threshold of 80%.

Maximum wind and PV capacities are limited to three times the peak demand (35 MW). Solving the full optimisation problem over a span of 365 days is infeasible. As such, five representative days are adopted with an hourly time interval using the scaled K-medoids clustering technique, capturing key variations in solar and wind generation, and load profiles. The choice of five representative days reflects a deliberate trade-off between computational tractability and degradation/operational accuracy. A systematic sensitivity analysis of representative-day-based battery degradation estimation by prior work [39] showed that increasing the number of representative days from five to ten or twenty does not necessarily improve degradation accuracy relative to a full-resolution “Base Case” simulation. The main reason is that representative days obtained via clustering primarily capture the most frequent (typical) operating patterns, but degradation-dominant extreme days with rare occurrence can remain under-represented regardless of cluster count. In that study, five representative days were found to offer a favourable balance between computational efficiency and accuracy, with additional days yielding diminishing and inconsistent accuracy gains while increasing runtime. Furthermore, the choice of the scaled K-medoids clustering technique, rather than alternative methods such as K-means or fuzzy clustering, is justified by prior work [17]. It was demonstrated that a scaled K-medoids approach more accurately preserves critical operational extremes, specifically peak load and maximum ambient temperature, relative to the real annual dataset, compared to K-means and fuzzy clustering methods that tend to smooth out these extremes. As these extremes are particularly important for sizing generation and storage systems and for assessing battery degradation under thermal stress, scaled K-medoids was adopted as the clustering technique in this study.

- The aggregated BESS capacity is limited to 35 MW/48 MWh, reflecting the technical envelope of commercially available containerised lithium-ion battery systems designed for high-power industrial and mining applications. The reserve delivery time is set to 2 seconds, with the primary reserve

contribution capped at 0.4 pu to represent short-duration high-power operation for fast frequency support. The BESS C-rate is constrained within the range of 0.25–2 h⁻¹, which is consistent with vendor specifications for mining-grade systems that provide high power-to-energy ratios for frequency control and spinning reserve. Moreover, the depreciation coefficients $\alpha_{dp} = 3.8837$ and $\beta_{dp} = 2.9702$ are adopted, corresponding to an average annual value reduction of 13.3% for lithium-ion batteries over a 10-year lifetime to reach 80% of initial energy capacity [40, 41].

- The aggregated diesel generator capacity cannot exceed 25 MW [17]. The inertia constant is 4 seconds with a reserve delivery time of 10 seconds, a governor deadband time of 0.5 seconds, a maximum governor ramp rate of 0.15 pu/s, and a primary reserve capped at 0.4 pu [42].

In this study, power disturbance is defined as a sudden 15% increase in total load [4]. The assumed 15% step load disturbance reflects the energisation of large underground mining assets such as fast-charging haul trucks, hoisting system, crushers, or conveyor start-ups, which are known to impose step changes exceeding 10% of site demand in practice [43, 44]. For instance, the newly commercialised electric mining truck [44], which is equipped with a 1.2 MWh battery and can be fully charged within approximately 50 minutes, corresponding to a charging power of about 1.44 MW. Accordingly, in a mine with an average demand of around 10 MW, a single fast-charging event introduces an additional load of approximately 1.5 MW. Such events may occur repeatedly throughout daily operations, thereby subjecting the MG to sustained frequency stress.

The nominal system frequency is set to $f^0 = 50$ Hz. In line with IEEE Std 1547-2018 and the Western Australia Electricity Market frequency standards, the reference RoCoF limit for inverter-interfaced resources in weak grids is 0.5 Hz/s [45, 46], and the normal operating frequency band for islanded systems is 49.5–50.5 Hz [47, 48]. However, based on technical consultations with power system and mining engineers at remote underground sites, these limits are frequently adapted in practice to accommodate large and frequent step-load variations associated with high-power mining equipment. Accordingly, slightly relaxed thresholds of $RoCoF^{\max} = 0.8$ Hz/s and $f^{\min} = 49.2$ Hz are adopted in this study to reflect realistic operating envelopes in remote mining MGs. To ensure maximum reliability, zero energy not served (ENS) is imposed, i.e., $\overline{ENS} = 0$.

The MISOCP optimisation problem was formulated in Pyomo and solved using the Gurobi solver [49, 50]. All numerical tests are performed on a high performance computing (HPC) system with Intel(R) Xeon(R) Platinum 8360Y CPU, clocking at 2.4GHz with 128 GB memory.

4.1. Numerical Results

This subsection presents the numerical results obtained by applying the proposed frequency-constrained MG planning model to the aforementioned mine site. The analysis is structured around three main themes: (i) the impact of frequency-related constraints, (ii) the effect of incorporating battery degradation modelling, and (iii) the influence of different mine lifespans.

Two primary planning scenarios are considered:

- Baseline (no PFR): Planning without any frequency constraints.
- Proposed (with PFR): Planning that incorporates frequency nadir, RoCoF, and steady-state frequency constraints.

These scenarios allow for a comprehensive comparison of the technical and financial implications of frequency support and battery ageing in MG planning.

4.1.1. Impact of frequency constraints

The optimal distributed energy resource (DER) configurations and associated cost breakdowns under different planning horizons are summarised in Tables 2 and 3, respectively. The results show that incorporating frequency constraints significantly influences the DER sizing strategy. Across all planning horizons, the proposed model selects a higher diesel generation capacity (13.5 MW) compared to the baseline scenario (12.0 MW). This increase reflects the need for fast and dispatchable power sources to support frequency stability. Moreover, the proposed model generally chooses smaller wind and solar systems. In the 15-year case, wind and solar capacities are reduced from 21.1 MW and 6.2 MW in the baseline to 18.1 MW and 5.6 MW with the proposed method, respectively. In particular, the baseline model avoids BESS in the 15- and 25-year scenarios, whereas the proposed model deploys BESS in all cases. For example, the 15-year proposed case includes a 5.9 MW/3.1 MWh BESS, highlighting the role of storage in maintaining frequency performance. This outcome may seem counterintuitive from a pure cost perspective, as diesel generators typically have lower upfront costs per kilowatt. However, frequency support constraints, specifically for RoCoF and frequency nadir, require fast and precise active power injection that diesel generators alone cannot sufficiently provide due to their slower ramp rates and inherent mechanical limitations. The BESS is therefore selected not as a cheaper solution, but as an enabler of improved dynamic system response, ensuring compliance with the strict frequency security requirements. This result underscores that even in cost-sensitive MG designs, the technical limitations of conventional assets can drive the inclusion of BESS to meet critical stability objectives.

The results also highlight a distinct difference in the BESS sizing strategies between the baseline and the proposed models. In the 7-year baseline case, which does not consider PFR constraints, the model selects a BESS with the same power and energy ratings (6.0 MW / 6.6 MWh), corresponding to a C-rate close to 1.0—indicating a general-purpose battery probably optimised for energy arbitrage. In contrast, the proposed model, which includes frequency stability constraints, consistently selects BESS configurations with high power (5.5–5.9 MW) and much lower energy capacities (3.1–3.2 MWh), resulting in a significantly higher C-rate. This reflects a strategic shift toward using BESS primarily for short-duration, high-power services such as RoCoF and frequency nadir support. The preference for high C-rate batteries becomes more pronounced as the planning horizon extends to 15 and 25 years, where degradation and replacement costs become increasingly influential. Energy-intensive batteries with high MWh ratings are more prone to deep cycling and accelerated wear, leading to higher lifecycle costs. By limiting the energy capacity while maintaining high power output, the proposed model reduces energy throughput per cycle and focuses the BESS role on transient frequency control. This sizing approach demonstrates the model’s ability to adapt battery configurations to technical needs and cost-effectiveness, especially in remote MGs that require frequency resilience.

Table 2: Optimal Configuration of MG Energy Resources

Scenario	Wind (MW)	Solar (MW)	Diesel (MW)	BESS	
				(MW)	(MWh)
Baseline (7yrs)	18.3	5.6	12.0	6.0	6.6
Proposed (7yrs)	18.1	5.6	13.5	5.9	3.1
Baseline (15yrs)	21.1	6.2	12.0	0.0	0.00
Proposed (15yrs)	18.1	5.6	13.5	5.9	3.1
Baseline (25yrs)	21.1	6.2	12.0	0.0	0.00
Proposed (25yrs)	19.9	6.5	13.5	5.5	3.2

Table 3: Planning Cost Breakdown for the Optimal Configurations

Scenario	C^I	C^O	C^R	C^S	C^E	NPC
	(\$M)	(\$M)	(\$M)	(\$M)	(\$M)	(\$M)
Baseline (7yrs)	70.48	32.52	0.00	31.24	6.56	78.32
Proposed (7yrs)	69.25	33.94	0.00	30.43	6.98	79.74
Baseline (15yrs)	71.86	62.90	0.00	13.01	12.26	133.99
Proposed (15yrs)	69.25	64.84	2.53	12.8	13.34	137.16
Baseline (25yrs)	71.86	91.39	0.00	4.17	17.81	176.88
Proposed (25yrs)	74.82	88.92	4.22	5.17	17.19	179.98

Introducing frequency constraints leads to a moderate increase in the total NPC. For the 15-year horizon, the NPC increases from \$133.99M in the baseline to \$137.16M under the proposed model (about 2.3% increase). Similarly, in the 25-year case, the NPC increases from \$176.88M to \$179.98M (approximately

1.7% increase). This additional cost is mainly driven by increased investment (C^I) and replacement costs (C^R) due to the deployment of BESS and higher diesel capacity. Specifically, in the proposed model, the BESS replacement cost is \$2.53M for the 15-year case and \$4.22M for the 25-year case, whereas it is absent in the baseline cases. In extended planning periods, the cost of replacing aged BESS units becomes a significant economic consideration, deterring their installation in baseline cases where frequency support was not required. However, the extra cost under the proposed model can be justified by meeting the frequency stability requirements.

To further investigate the impact of frequency-related constraints on the MG design outcomes, the frequency responses of the optimal configurations listed in Table 2 were analysed through time-domain simulations conducted in MATLAB/Simulink. The simulated MG adopts a droop-based frequency response control structure in which the BESS provides fast converter-based active power support immediately following a disturbance, while the diesel generators contribute through slower governor-based droop control. Figure 4 contains representative time-domain frequency response curves obtained from the MATLAB/Simulink simulations for two distinct scenarios. Scenario 1 corresponds to a representative disturbance interval from the 15-year and 25-year planning cases, in which no BESS was installed in baseline configurations and frequency support was provided entirely by diesel generators. Scenario 2 corresponds to a representative disturbance interval from the 7-year planning case, in which BESS was installed in the proposed and conventional configurations. However, the conventional (baseline) system's BESS had depleted its energy at the time of the disturbance, and the lower online diesel generation capacity was insufficient to respond to the disturbance. In both scenarios, the proposed method maintains the frequency nadir well above the 49.2 Hz limit (49.821 Hz and 49.832 Hz, respectively). In contrast, the conventional baseline violates the frequency nadir limit in both scenarios, reaching nadirs of 48.229 Hz and 48.566 Hz, respectively. Notably, in Scenario 1, the baseline system also violates the RoCoF limit, exhibiting a post-disturbance RoCoF of approximately 1.55 Hz/s, well above the prescribed $RoCoF^{max} = 0.8$ Hz/s threshold, due to the absence of any fast-acting BESS response to arrest the initial rate of frequency decline. In Scenario 2, although the post-disturbance RoCoF of the baseline system (0.31 Hz/s) remains within the prescribed limit, the inadequate primary reserve resulting from the BESS's depleted energy reserve at the disturbance instant leads to a substantially deeper and slower-recovering frequency excursion that violates the nadir limit. These results provide direct transient-domain evidence that the proposed frequency-constrained planning framework ensures compliance with both RoCoF and frequency nadir security requirements, whereas the conventional approach can violate one or both of these requirements, particularly in the absence of adequate fast-acting BESS support.

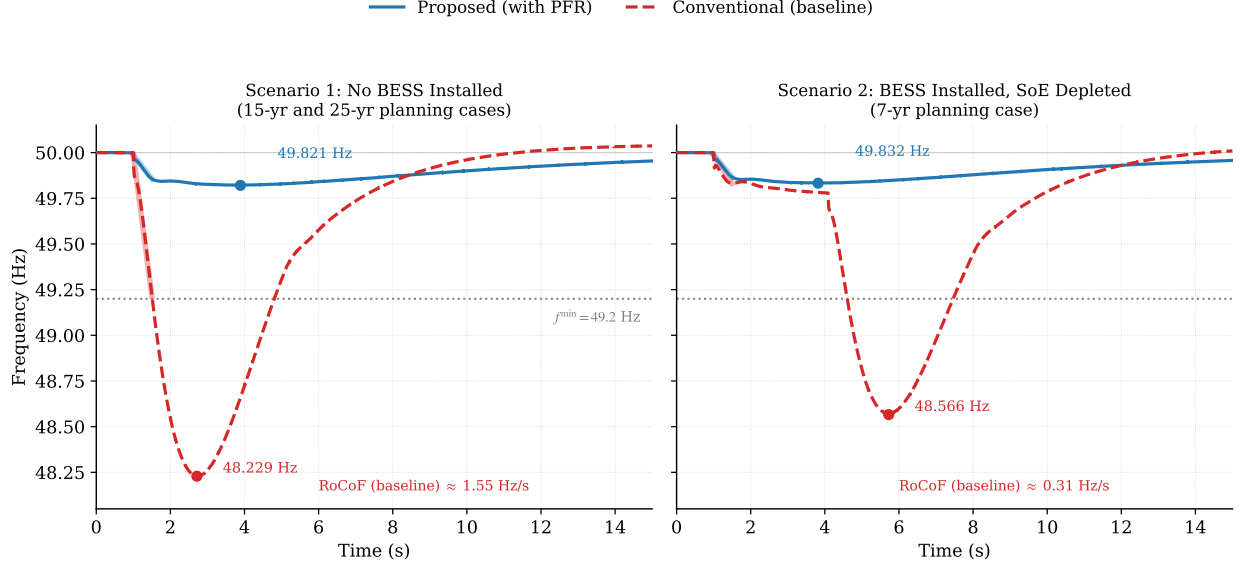


Figure 4: Representative time-domain frequency response following a disturbance for the proposed and conventional approaches: (a) Scenario 1, no BESS installed (15-yr and 25-yr cases); (b) Scenario 2, BESS installed but depleted at the disturbance instant in the conventional case (7-yr case).

Moreover, Figure 5 aggregates transient behaviour modelled by MATLAB/Simulink and illustrates the changes in RoCoF and frequency nadir over all time intervals for the proposed and baseline approaches. The results indicate that the proposed method successfully maintains the frequency nadir and RoCoF within acceptable limits throughout the planning period, specifically $RoCoF^{\max} = 0.8 \text{ Hz/s}$ and $f^{\min} = 49.2 \text{ Hz}$. By contrast, the baseline method frequently violates both the RoCoF and frequency nadir limits, with more than 40% of the operating intervals exceeding the allowable thresholds. This behaviour is primarily attributed to the reduced diesel generator capacity in the baseline planning cases, which results in lower aggregated system inertia and limited governor-based droop response capability. For example, in the 7-year planning study, which includes a 6.0 MW/6.6 MWh BESS, poor battery operation for frequency support and a relatively smaller diesel generator capacity cause RoCoF and frequency limit violations in approximately 43% of the scenarios. This is because, following a disturbance, insufficient inertial and droop-based frequency support is available during the critical first seconds, leading to steeper initial frequency gradients that push RoCoF and nadir values outside acceptable limits. These findings clearly demonstrate that neglecting frequency security constraints at the planning stage leads to under-sized synchronous generators and exposes the MG to unacceptable dynamic performance, highlighting the necessity of embedding frequency stability considerations directly into the MG design framework.

Moreover, the baseline approach results in significant deviations from frequency stability metrics. No-

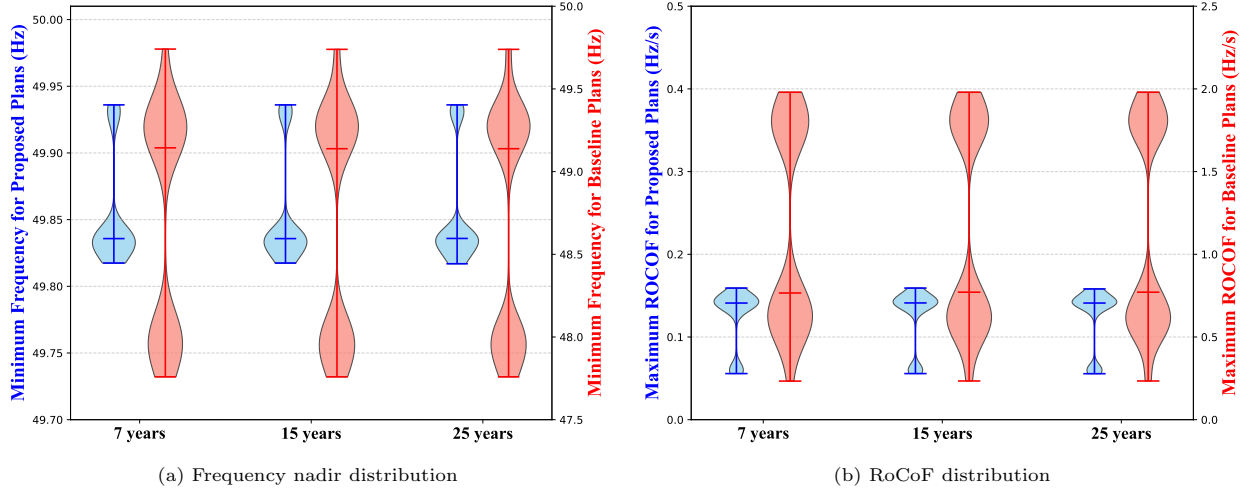


Figure 5: Post-disturbance frequency response under the proposed method (black, left y-axis) and the baseline method (red, right y-axis) for all cases.

tably, the frequency nadir falls to 47.7 Hz—far below the standard threshold of 49.2 Hz. Likewise, RoCoF peaks at 1.98 Hz/s within the first second following a disturbance, significantly exceeding the acceptable limit.

4.1.2. Battery degradation assessment

The proposed model incorporates a linearised representation of battery degradation directly into the optimisation problem, enabling a degradation-aware MG planning framework. To assess the fidelity of this linear approximation, a post-optimisation validation is performed by applying a detailed degradation model to the optimised SOC profiles. This post-validation uses: (i) Rainflow counting to extract cycle depths from the SOC profiles, and (ii) Non-linear degradation equations (25) and (26), which are based on experimental data. The goal is to evaluate how well the linear degradation approximation aligns with real-world degradation behaviour and to quantify any mismatch in capacity fade and replacement cost.

Battery degradation is affected by multiple interacting stress factors including DoD, C-rate, average SoC and temperature. However, as reported in [28, 30], when the C-rate is below unity and temperature is approximately regulated, cycle and calendar ageing can be reliably approximated using DoD- and SoC-dependent terms, respectively, while the explicit C-rate contribution can be neglected with limited loss of accuracy. This operating regime is typical for stationary BESS in MG applications.

To verify that this assumption holds for the considered case studies, Figure 6 presents a box plot of the operational C-rate distribution of the BESS across all analysed scenarios. As shown in this figure, the median C-rate in all cases remains well below 1, with the vast majority of operating points lying in the

sub-1C region. This confirms that the battery is rarely subjected to high-stress charge or discharge events, justifying the adopted degradation formulation.

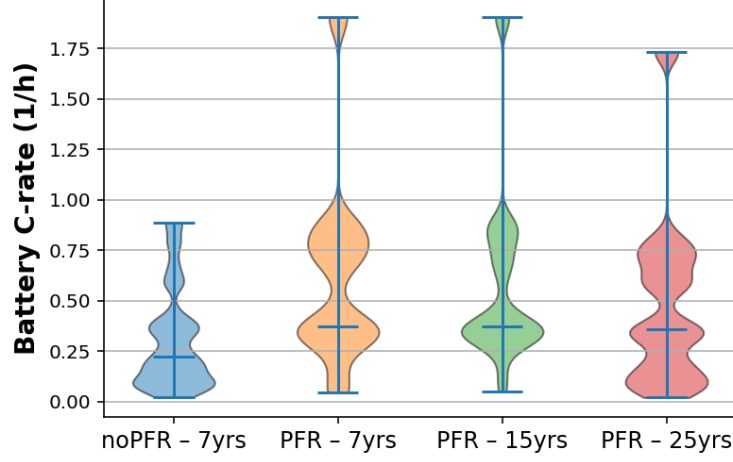


Figure 6: Box plot of BESS operational C-rate for the four scenarios with installed storage (noPFR-7yrs, PFR-7yrs, PFR-15yrs and PFR-25yrs).

Of the six scenarios analysed (comprising three lifetimes and two sizing approaches), those without the incorporation of BESS have been excluded from this analysis. Figure 7 presents the remaining battery capacity over the planning years for each scenario, using both the linearised degradation model used in the optimisation and the post-processed non-linear model.

The comparison between the proposed linearised degradation model and the non-linear benchmark reveals that the linear model closely tracks the non-linear degradation trajectory in most scenarios. Specifically, for the noPFR-7yrs, PFR-7yrs, and PFR-25yrs cases, the replacement timings under the linear and non-linear models are closely aligned, typically within a one-year margin, demonstrating that the proposed model provides a reliable approximation for long-term planning. The only noticeable deviation occurs in the PFR-15yrs scenario, where the linear model predicts the need for battery replacement at year 8, while the non-linear benchmark delays this to year 11.

This deviation stems from the more aggressive cycling profile in the PFR-15yrs case. In this scenario, the BESS is tasked with a frequent and rapid response to satisfy RoCoF and frequency nadir constraints, which amplifies the effect of cumulative degradation. The linear model, by design, aggregates the degradation effects more conservatively, leading to earlier replacement triggers. Notably, this bias works in favour of planning security: model errors on the side of caution. In practice, this means that the actual cost of ownership could be lower than the optimisation output, an outcome generally welcomed in capital-intensive applications where asset reliability is critical.

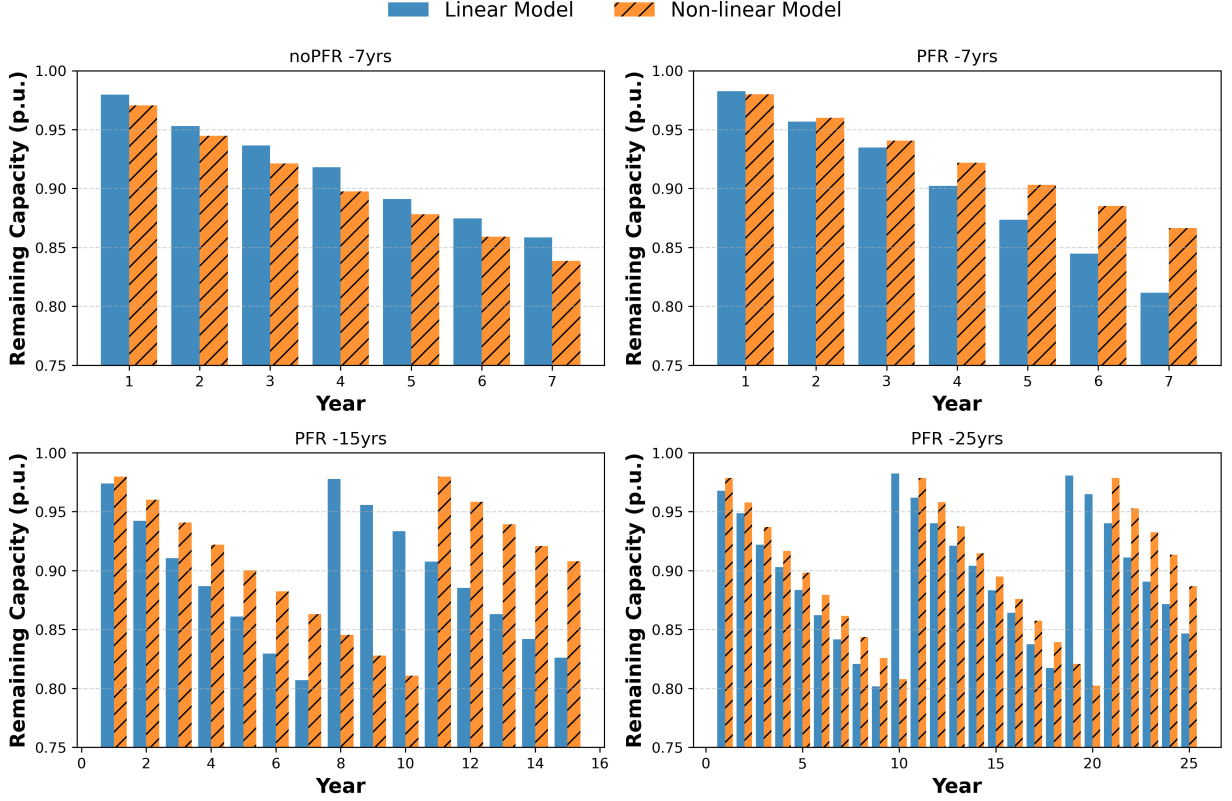


Figure 7: Comparison of battery capacity fade over the planning years for four scenarios (noPFR-7yrs, PFR-7yrs, PFR-15yrs, and PFR-25yrs). Solid bars represent degradation modelled using the linearised approach embedded in the optimisation, while hatched bars show post-processed non-linear degradation based on actual SoC profiles.

While the accuracy of degradation modelling is important, it is equally essential to evaluate its financial implications. The proposed linearised model results in an overestimation of total replacement costs by 5.9% and 2.8% in the PFR-15yr and PFR-25yr cases, respectively. Although conservative, the proposed model yields financially robust outcomes that are well suited for long-term planning and are unlikely to impact the financial viability or bankability of the project.

4.2. Sensitivity Analysis

To verify the robustness of the proposed frequency-constrained planning framework, a sensitivity analysis is conducted across three dimensions: (i) demand growth rate, (ii) frequency operating standards (RoCoF limit and frequency nadir threshold), and (iii) alternative battery degradation parameter sets. In all cases the 15-year planning horizon is used as the reference, since it captures meaningful degradation dynamics while keeping the computational burden tractable across the full sweep of scenarios. For each experiment only one group of parameters is varied while the remaining parameters are held at their base-case values (base-case demand, $\text{RoCoF}^{\max} = 0.8 \text{ Hz/s}$, $\Delta f^{\max} = 0.8 \text{ Hz}$, baseline NMC degradation parameters).

4.2.1. Demand Growth Rate

Remote mining operations typically experience significant demand evolution over their lifetime, driven by fleet electrification, ore grade decline, and production expansion. To assess the sensitivity of the proposed framework to such trends, the base-case demand profile is uniformly scaled to represent demand growth rates of -20% , -10% , $+10\%$, and $+20\%$ relative to the reference level, in addition to the base case. Table 4 summarises the optimal DER sizing and total NPC for both the proposed (PFR) and baseline (no-PFR) models under each demand level.

Table 4: Sensitivity to demand growth rate (15-year horizon, NMC, $\text{RoCoF}^{\max} = 0.8 \text{ Hz/s}$, $\Delta f^{\max} = 0.8 \text{ Hz}$).

Demand deviation	Scenario	Wind (MW)	Solar (MW)	Diesel (MW)	BESS (MW/MWh)	C^I (\$M)	C^O (\$M)	C^R (\$M)	C^S (\$M)	NPC (\$M)
-20%	Baseline	16.9	4.9	10.0	— / —	57.96	50.46	0.00	10.50	107.77
	Proposed	16.0	5.2	10.8	3.4 / 1.7	59.20	50.04	3.23	11.13	111.11
-10%	Baseline	19.0	5.6	11.0	— / —	64.90	56.67	0.00	11.75	120.87
	Proposed	18.0	5.8	12.1	4.5 / 3.2	67.64	55.31	3.94	13.36	124.23
Base	Baseline	21.1	6.2	12.0	— / —	71.86	62.90	0.00	13.01	133.99
	Proposed	18.1	5.6	13.5	5.9 / 3.1	69.25	64.84	2.53	12.80	137.16
$+10\%$	Baseline	23.2	6.8	13.0	— / —	78.79	69.09	0.00	14.27	147.07
	Proposed	22.1	7.1	14.8	4.7 / 2.4	81.62	67.97	3.12	15.40	150.50
$+20\%$	Baseline	25.3	7.4	14.0	— / —	85.74	75.31	0.00	15.53	160.18
	Proposed	24.2	7.7	16.1	5.1 / 2.5	89.08	74.10	6.42	16.82	167.15

As can be seen, the baseline never deploys BESS, regardless of demand level. The proposed model, on the other hand, includes storage in every case — the frequency constraints simply cannot be met without it, no matter how small or large the load. This is perhaps the clearest illustration of why frequency-aware planning matters: a design that looks adequate from a cost perspective may be quietly non-compliant in practice.

Beyond storage, the proposed model also requires more diesel than the baseline at every demand level. At -20% demand the gap is modest (10.8 MW vs. 10.0 MW), but widens to 2.1 MW at $+20\%$ (16.1 MW vs. 14.0 MW). The reason is straightforward: larger loads mean larger disturbances (defined here as 15% of total demand), which demand more inertia and faster reserve delivery. The combined effect of higher diesel and BESS investment produces an NPC premium over the baseline that grows from \$3.3M at -20% to \$7.0M at $+20\%$. Given that these costs buy guaranteed frequency compliance — something the baseline cannot offer — the premium appears well justified.

One cost item worth noting is the replacement cost C^R , which jumps to \$6.42M at $+20\%$ demand

compared with \$2.53M at the base case. More intensive load disturbances accelerate battery cycling and shorten the replacement interval. The baseline avoids this cost entirely, but only because it carries no storage at all. Across all demand levels tested, the proposed model remains frequency-compliant and adapts sensibly to changing load, confirming that the framework is robust to realistic demand uncertainty.

4.2.2. Frequency Operating Standards

Grid frequency standards—specifically the maximum allowable RoCoF and the minimum frequency nadir—are key regulatory inputs that directly govern the stringency of the frequency security constraints embedded in the proposed model. To assess the sensitivity of planning outcomes to these thresholds, four settings are evaluated: a strict standard ($\text{RoCoF}^{\max} = 0.5 \text{ Hz/s}$, $\Delta f^{\max} = 0.5 \text{ Hz}$), the base case ($0.8 \text{ Hz/s} / 0.8 \text{ Hz}$), and two relaxed standards ($1.0 \text{ Hz/s} / 1.0 \text{ Hz}$ and $1.5 \text{ Hz/s} / 1.5 \text{ Hz}$), broadly reflecting the range found in international grid codes for islanded and weak-grid systems [47]. The results are summarised in Table 5.

Table 5: Sensitivity to frequency operating standards (15-year horizon, base-case demand, NMC).

$\text{RoCoF}^{\max} / \Delta f^{\max}$ (Hz/s / Hz)	Wind (MW)	Solar (MW)	Diesel (MW)	BESS (MW/MWh)	C^I (\$M)	C^O (\$M)	C^R (\$M)	C^S (\$M)	NPC (\$M)
0.5 / 0.5 (strict)	19.9	6.5	21.5	4.4 / 3.2	84.18	61.76	3.83	16.57	145.22
0.8 / 0.8 (base)	18.1	5.6	13.5	5.9 / 3.1	69.25	64.84	2.53	12.80	137.16
1.0 / 1.0 (relaxed)	19.9	6.5	12.0	3.9 / 2.2	71.98	62.69	1.82	13.22	135.57
1.5 / 1.5 (relaxed)	20.3	6.4	12.0	0.9 / 1.2	71.50	63.86	0.74	13.23	135.40

The results reveal several important insights. First, tightening the frequency standard to $\text{RoCoF}^{\max} = 0.5 \text{ Hz/s}$ and $\Delta f^{\max} = 0.5 \text{ Hz}$ leads to a substantial increase in diesel capacity (21.5 MW vs. 13.5 MW at base) to provide the higher inertia required by the stricter RoCoF constraint. BESS is still deployed (4.4 MW/3.2 MWh) to assist with the fast PFR requirement, and the overall NPC rises to \$145.22M—an increase of \$8.06M (5.9%) over the base case. The higher capital cost C^I (\$84.18M vs. \$69.25M) is the primary driver, reflecting the significant expansion of diesel capacity needed to satisfy the tight inertia requirement.

Second, as the frequency standard is progressively relaxed from the base case to 1.0/1.0 and 1.5/1.5, diesel capacity drops to its minimum bound of 12.0 MW while BESS power and energy ratings decrease substantially—from 5.9 MW/3.1 MWh at base to 3.9 MW/2.2 MWh at 1.0/1.0 and further to 0.9 MW/1.2 MWh at 1.5/1.5. This reflects the diminishing PFR obligation as standards are relaxed, allowing the model to reduce storage investment while maintaining feasibility. Notably, BESS remains deployed even at the most

relaxed standard, albeit at a minimal level, indicating that a small storage capacity is still economically justified to complement diesel PFR even under lenient frequency requirements.

Third, the NPC decreases monotonically as standards are relaxed (\$145.22M $\rightarrow$ \$137.16M $\rightarrow$ \$135.57M $\rightarrow$ \$135.40M), with the sharpest reduction occurring between the strict and base-case settings (\$8.06M), while the difference between the two relaxed cases is marginal (\$0.17M), suggesting that NPC converges to a lower bound beyond approximately 1.0 Hz/s. The replacement cost C^R follows the same trend, decreasing from \$3.83M at 0.5/0.5 to \$0.74M at 1.5/1.5, directly reflecting reduced BESS cycling intensity. Overall, this analysis demonstrates that the proposed framework responds predictably and consistently to variations in frequency standards, with NPC increasing monotonically as standards tighten. The results confirm that the base-case setting of 0.8/0.8 Hz/s represents a well-balanced operating point between frequency security and cost, and that significantly stricter standards impose a meaningful cost penalty primarily through expanded diesel capacity, offering practical guidance for regulators and mine operators when setting site-specific frequency requirements.

4.2.3. Sensitivity to Degradation Parameter Sets

This test asks a different question from the previous two: rather than changing what the system looks like, it examines what happens to cost and replacement scheduling if the battery degrades differently from what was assumed at the planning stage. To investigate this, the optimal MG configuration from the PFR-15yrs case (Table 2) is held fixed, and only the degradation model coefficients are swapped — first for those of lithium iron phosphate (LFP), then for lithium manganese oxide (LMO), both taken from the same experimental source used to calibrate the baseline NMC parameters [28]. LFP is known for its long cycle life and thermal stability; LMO tends to degrade more quickly, particularly under deep cycling. Results are shown in Table 6.

Table 6: Sensitivity analysis using alternative degradation parameter sets (15-year horizon, base-case demand, $\text{RoCoF}^{\max} = 0.8 \text{ Hz/s}$, $\Delta f^{\max} = 0.8 \text{ Hz}$).

Parameter set	BESS (MW/MWh)	Replacement year(s)	C^R (\$M)	NPC (\$M)	Freq. compliant
NMC (base case)	5.9 / 3.1	12	2.53	137.16	Yes
LFP parameters	5.9 / 3.1	—	—	136.17	Yes
LMO parameters	5.9 / 3.1	5 and 9	4.33	139.37	Yes

The results are intuitive but the magnitudes are instructive. With LFP parameters, the battery never reaches its end-of-life threshold over 15 years, so no replacement is needed and the NPC drops by \$0.99M relative to the NMC baseline. With LMO parameters, faster degradation forces two replacements — at years

5 and 9 — pushing C^R to \$4.33M and lifting the NPC to \$139.37M, a \$2.21M increase. The total spread across the three parameter sets is just over \$3.2M, which is not trivial for a capital-intensive remote project. Frequency compliance is maintained in all three cases, confirming that the planned system configuration is operationally robust to degradation uncertainty, even if the associated cost is not.

What this test highlights more broadly is that degradation modelling is not merely a technical detail — it has real financial consequences. A planner who assumes optimistic LFP-like degradation when the installed cells behave more like LMO could face unexpected replacement costs mid-project. The framework proposed here, by embedding degradation directly into the optimisation, ensures that whatever parameters are used are treated consistently and transparently throughout the planning process.

4.2.4. Summary of Sensitivity Analysis

Table 7 provides a consolidated overview of the directional impact of each sensitivity dimension on the key techno-economic planning outcomes of the proposed PFR-constrained framework.

Table 7: Summary of sensitivity analysis: directional impact on key metrics under the proposed PFR-constrained model. (↑ increase, ↓ decrease, ↔ negligible change).

Parameter variation	Diesel capacity	BESS power	NPC	Freq. compliance
Demand growth ↑	↑	↑	↑	Maintained
Demand growth ↓	↓	↓	↓	Maintained
Stricter freq. standard	↑	↑	↑	Maintained
Relaxed freq. standard	↓	↓	↓	Maintained
LFP parameters (vs. NMC)	↔	↔	↓	Maintained
LMO parameters (vs. NMC)	↔	↔	↑	Maintained

Looking across all three sensitivity dimensions, a consistent picture emerges. Demand and frequency standards are the primary drivers of system sizing: higher loads or tighter thresholds push the model toward more diesel and more storage, with NPC rising accordingly. The trends are monotonic and physically sensible, which builds confidence that the formulation is behaving correctly. Across every feasible scenario tested, frequency compliance is maintained — the proposed framework does not trade dynamic security for cost savings.

The most practically significant finding concerns the role of diesel under strict frequency standards. Tighter limits increase system cost primarily by adding diesel rather than storage, because RoCoF compliance is driven by synchronous inertia. For mine operators and regulators, this is an important nuance: adopting very strict frequency standards in a remote, low-inertia MG may inadvertently lock in a larger fossil-fuel footprint, working against the decarbonisation goals that typically motivate renewable integration

in the first place.

On the degradation side, swapping parameter sets shifts total lifecycle cost by more than \$3M over 15 years without changing a single investment decision. This underscores the practical value of embedding degradation directly into the optimisation, as done here, rather than treating it as a post-processing correction. Taken together, the sensitivity results confirm that the proposed framework is robust across a realistic range of operating conditions and modelling assumptions, and that its outputs are physically consistent and practically interpretable.

4.3. Discussion on Battery Degradation Modelling Assumptions

The degradation model adopted in this study (Section 2) estimates cycle ageing from the number and depth of charge-discharge cycles, and calendar ageing from the average SoC and elapsed time, holding temperature, C-rate, and dwell time at representative baseline conditions. To assess whether these additional factors could, in principle, be incorporated into the quadratic degradation functions used in this study, a supplementary calibration exercise was carried out using a reference NMC lithium-ion cell ageing model (BLAST-Lite Kokam75Ah [33]). For each representative operating profile, one-year calendar and cyclic capacity-loss trajectories were simulated and converted to per-day and per-cycle rates — calendar ageing per day (as a function of average SoC, at DoD = 10%) and cyclic ageing per cycle (as a function of DoD, at average SoC = 50%). Then, quadratic functions of the same form used in this study were re-fitted to each. Figure 8 summarises the resulting curves across four ambient temperatures (10–40°C), three C-rates (0.5–3C), and four dwell (idle) times (0–18 h per cycle).

Temperature was confirmed as the dominant driver of both calendar and cyclic ageing on a per-day and per-cycle basis (middle column in Figure 8), with both coefficients increasing noticeably and monotonically with temperature, consistent with Arrhenius-type behaviour [51]. This supports the assumption of temperature-regulated BESS enclosures adopted in this study, because even a moderate rise in operating temperature would otherwise substantially increase the per-day and per-cycle degradation rate.

C-rate was found to have no meaningful direct effect on calendar ageing (Figure 8(a)), and only a weak, second-order effect on cyclic ageing once expressed on a per-cycle basis (Figure 8(d)). This confirms that the apparent C-rate effect arises predominantly from the number of charge-discharge cycles completed at higher C-rates, rather than from a direct per-cycle degradation mechanism. Because the degradation model adopted in this study already explicitly counts the number and depth of charge-discharge cycles, rather than assuming a fixed throughput, this indirect C-rate effect is inherently captured by the existing formulation.

Dwell time was found to have a negligible effect on calendar ageing on a per-day basis (Figure 8(c)), resolving the open question identified in earlier stages of this calibration exercise. However, dwell time was found to have a genuine, direct effect on cyclic ageing (Figure 8(f)). Unlike C-rate, increasing dwell time raises the per-cycle degradation rate by approximately 38% between 0 h and 18 h of idle time per cycle, indicating that each individual cycle causes more damage when bookended by longer rest periods, independent of the reduction in the number of cycles completed. This finding suggests that dwell time is a genuine, second-order degradation mechanism not captured by the present cycle-counting formulation, and represents a specific, well-defined target for future model refinement.

To further examine the interaction between the two averaging dimensions used in Figure 8 (DoD for cyclic ageing, average SoC for calendar ageing), Figure 9 presents calendar ageing re-fitted across a range of DoD values, and cyclic ageing re-fitted across a range of average SoC values. Figure 9(a) shows that calendar ageing does depend on DoD, with the calendar-ageing-avg-SoC curve flattening as DoD increases from 0% to 90%. This happens because a wider cycling swing causes the cell to spend time away from its nominal average SoC at both higher and lower extremes, and calendar ageing is a convex function of instantaneous SoC. Figure 9(b) shows that cyclic ageing is, to the precision of this calibration, independent of where the DoD window is centred on the SoC axis: all ten avg-SoC curves (0–90%) collapse onto a single curve. This is consistent with the underlying cell model, in which the dominant cyclic ageing mechanism depends only on temperature and DoD, while the secondary avg-SoC-dependent term contributes negligibly to this regime.

It should be noted that the coefficients obtained through this calibration exercise are not directly comparable in scale to those of [28] adopted in this study, as the latter are formulated as rate constants intended to combine with equivalent full cycle (EFC) terms, rather than as independently fitted per-day and per-cycle coefficients. It is also noted that the per-day and per-cycle rates presented here represent trajectory, i.e., averaged values over a full year rather than instantaneous or marginal rates, since both the calendar and cyclic ageing trajectories in the reference model follow non-linear (sub-linear and sigmoidal, respectively) paths over time. Reconciling the two coefficient sets into a single, consistent, temperature-, C-rate-, and dwell-resolved degradation model compatible with the present formulation, and explicitly embedding the confirmed dwell-time effect using the duty-cycle information already available for each representative day, is identified as a valuable direction for future work.

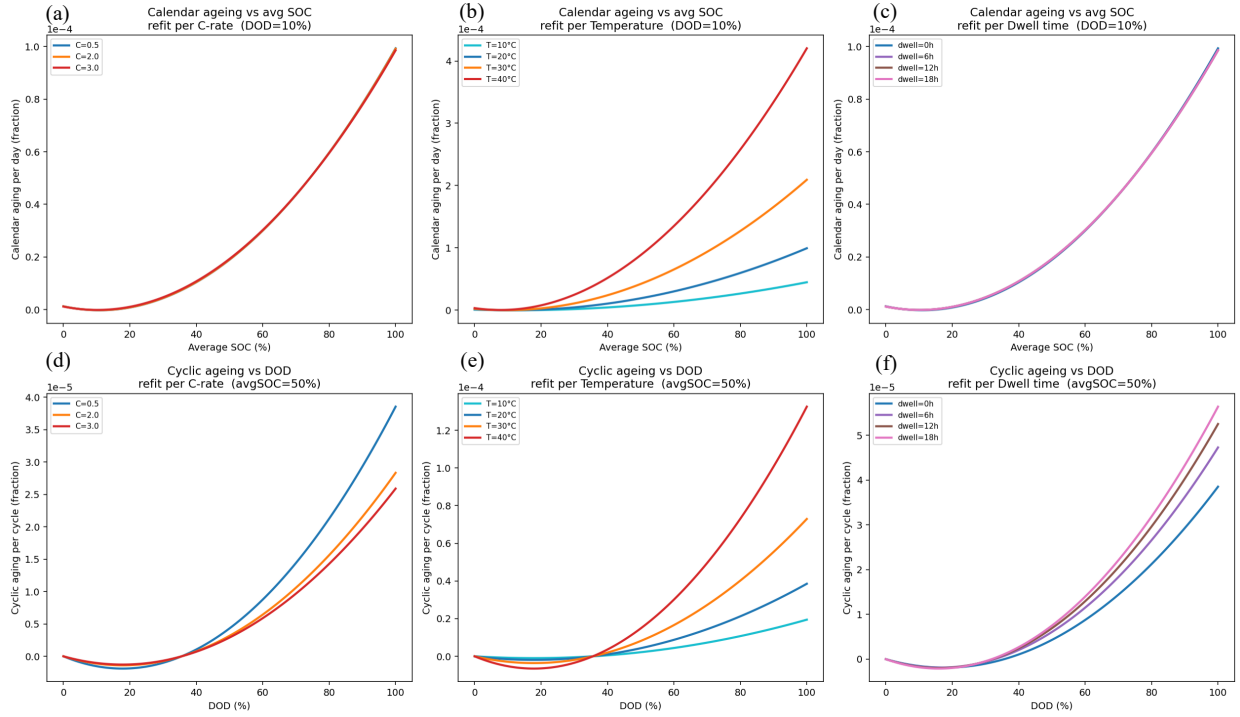


Figure 8: Effect of C-rate, temperature, and dwell time on calendar ageing per day (top row, vs. average SoC) and cyclic ageing per cycle (bottom row, vs. DoD), obtained by re-fitting the quadratic degradation functional form used in this study against a reference NMC cell model.

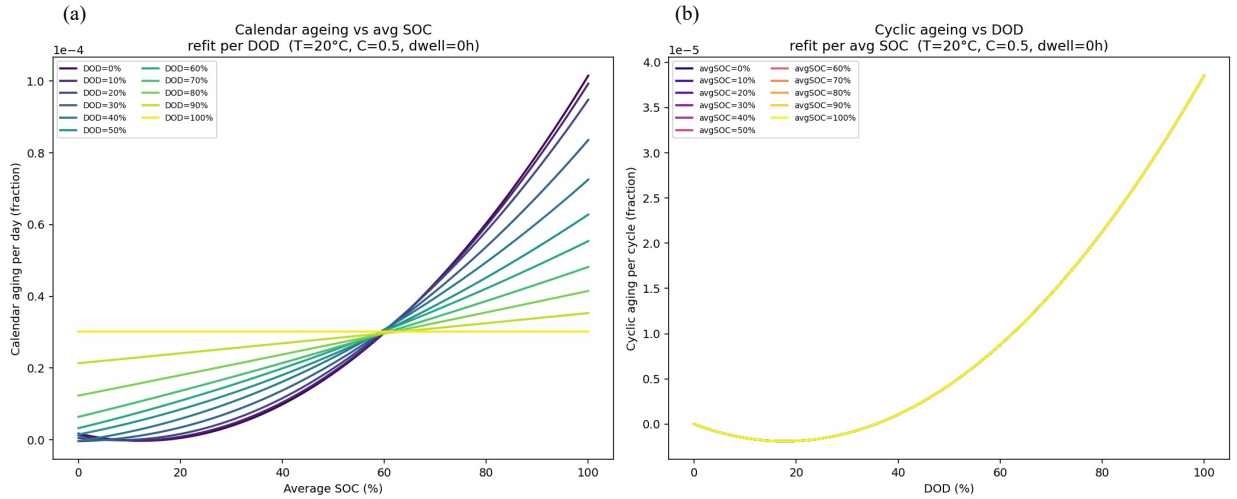


Figure 9: Interaction between the two averaging dimensions: (a) calendar ageing per day vs. average SoC, re-fitted across DoD values; (b) cyclic ageing per cycle vs. DoD, re-fitted across average SoC values.

5. Conclusions

This paper introduces an innovative MG planning framework with frequency constraints and degradation modelling, specifically tailored for low-inertia industrial applications like remote mining operations. The model incorporates frequency stability constraints—including RoCoF, frequency nadir, and steady-state limits—within a MISOCP framework to optimally size RES, lithium-ion BESS, and diesel generators, minimising total NPC. The numerical results show that including frequency constraints leads to increased diesel capacity (from 12.00 MW to 13.46 MW) and BESS deployment in all proposed cases, resulting in a moderate increase in NPC of up to \$3.1M over 25 years. The proposed model maintains frequency within limits (minimum 49.2 Hz, RoCoF below 0.8 Hz/s), whereas the baseline violates these limits in more than 40% of scenarios.

A key contribution is the integration of a linearised battery degradation model that captures calendar and cycling effects through SOS constraints, thereby improving the reliability and economic efficiency of battery management. The model predicts earlier battery replacements than a non-linear benchmark, increasing replacement costs by 5.92% and 2.84% for 15- and 25-year horizons, respectively. Despite conservative estimates, this approach allows more realistic planning and supports long-term MG stability.

Limitations and Future Work

Although this study provides new insights into frequency- and degradation-aware planning of mining MGs, several limitations remain. First, the uncertainty of renewable generation and demand was not explicitly modelled using stochastic or robust approaches to maintain tractability of the optimisation problem and focus on the impacts of frequency requirements and battery degradation. High-resolution RES and load data were used to identify representative days, enabling accurate capture of the dynamic effects of frequency constraints. Future work will extend the framework to incorporate stochastic or robust formulations, allowing variability in RES output and demand to be considered. Second, conservative transmission-level frequency limits were adopted; system-specific limits for mining MGs should be derived and validated in future studies. Third, the model focused on lithium-ion batteries, while emerging storage technologies such as flow batteries and hydrogen-based systems could be incorporated in subsequent work. Fourth, a single fixed calendar and cyclic ageing coefficient set was adopted throughout the planning horizon, rather than coefficients that vary with temperature, C-rate, or dwell time; future work will embed temperature-dependent coefficients using the ambient temperature data, and extend the model to capture C-rate and dwell-time effects.

Future research will also extend the model to include voltage and transient stability constraints and

improve degradation modelling accuracy through a post-optimisation validation step using a non-linear degradation model to refine lifecycle cost estimates. Hybrid degradation approaches, such as adaptive piecewise-linear models, could also be considered. These enhancements aim to advance sustainable and reliable MG solutions for remote heavy industry applications. Developing a linearised artificial neural network (ANN)-based battery ageing model and embedding it directly within the optimisation framework will also be considered as a future research direction. This will enable explicit representation of C-rate-dependent degradation while retaining computational tractability for long-term planning studies.

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